HW 2 - Grading - rubrics (by JK, 2019-10-11)

- Part 1: Loading data in NumPy (7 points)

  - **Question 1 (1 point):** ½ point for correctly included nump/matplotlib
  - **Question 2 (1 point):** successfully loaded and partially printed numpy array
  - **Question 3 (3 points):** 1 ppoint for each anwser:
    - How many separate children were measured in this data?
    - What is the age of the youngest and oldest child in this dataset?
    - Likewise, what is the height of the tallest child?
  - **Question 4 (2 point):** 1 point for each answer:
    - average waist
    - standard deviation

- Part 2: Making plots with Matplotlib (10 points)

  - **Question 5 (2 point):** plot of histogram with labels and title
  - **Question 6 (1 point):** any answer between 9 and 15 is OK
  - **Question 7 (4 points):** 2 points for each plot - one for scatter plot with prropriate markr size, another for labels/title
  - **Question 8 (1 point):** This is a loaded question. Acceptable answers include, initially for younger children height increases more than weight.
    After about 10 years of age, the opposite could be said or both increase similarly.
  - **Question 9 (2 point):** 1 points for 2D plot with color bar, 1 for labeling both plot and colorbar

- Part 3: Working with data in NumPy (11 points)

  - **Question 10 (2 points):** 1 point for corrent equation, 1 to accept correct parameters and return single value
  - **Question 11 (4 points):** 1 point for each of the following:

    ```
    - while loop over all data and
    - create list of BMI values by calling before created function
    - save into file
    - plot data
    ```

  - **Question 12 (2 points):** 1 point for loading the data, 1 point for plotting the data
  - **Question 13 (3 points):** 1 point for peopel over 18, 1 point for all with BMI > 25, 1 point for printing correct number of people

- Part 4: Modeling Growth (12 points)

  - **Question 14 (1 points):** create a function according to equation
  - **Question 15 (4 points):**
    - 1 point for `increase_age`
    - 1 point for computing `linear_height`
    - 2 points for plot (probably: 1 for both series, 1 for labeling the plot)
  - **Question 16 (2 points):** 1 point for each function - accepting 'age' and returning 'height'
  - **Question 17 (4 points):** 2 points for correctly computing values, 2 points for plotting (probably: 1 for both series, 1 for labeling)
  - **Question 18 (1 points):** 1 point for 'pointing out that ICP is better a early and later age - as it fits better'